



GI Infections & Viral Hepatitis Therapy — MCQ Practice

Pharmacology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. For most viral GI infections (e.g. rotavirus, norovirus), what is the recommended treatment approach?

EASY

- A. Specific antiviral drug therapy
- B. Conservative management only — antipyretics, antiemetics, oral rehydration
- C. Broad-spectrum antibiotics
- D. Immediate hospitalization for all cases
- E. Antifungal therapy

2. What is the WHO-recommended first-line antibiotic for Shigella (bacillary dysentery)?

EASY

- A. Metronidazole
- B. Ciprofloxacin
- C. Vancomycin
- D. Doxycycline
- E. Azithromycin only

3. What is the first step in managing suspected C. difficile colitis?

EASY

- A. Start vancomycin immediately
- B. Stop the causative/offending antibiotic
- C. Add loperamide
- D. Start ciprofloxacin
- E. Perform immediate colectomy

4. Which antiviral drug class is used for BOTH chronic HBV and HCV treatment?

EASY

- A. Nucleoside analogues (lamivudine, entecavir)
- B. Interferon-alpha (pegylated)
- C. Sofosbuvir
- D. Ribavirin alone
- E. Protease inhibitors alone

5. A pregnant woman develops V. cholerae infection. Which antibiotic is the safest choice?

MEDIUM

- A. Doxycycline
- B. Ciprofloxacin
- C. Azithromycin
- D. Tetracycline
- E. None are safe in pregnancy

6. A patient develops watery diarrhea 5 days after starting clindamycin for a dental abscess, with a positive C. difficile toxin assay. What is the correct sequence of management?

MEDIUM

- A. Start vancomycin immediately without stopping clindamycin
- B. Stop the clindamycin first, then start oral metronidazole (first choice), reserving oral vancomycin for treatment failure
- C. Add loperamide to control symptoms as the sole intervention
- D. Start ciprofloxacin instead
- E. No action needed — this will resolve spontaneously without any intervention

7. A patient develops non-typhoid Salmonella gastroenteritis and is 10 years old. Ciprofloxacin is being considered. What is the concern, and what is the alternative?

MEDIUM

- A. Ciprofloxacin has no age restriction and is the correct choice
- B. Ciprofloxacin is contraindicated under 14 years; ceftriaxone is the safe alternative in this age group
- C. Ceftriaxone is contraindicated in children, making ciprofloxacin the only choice
- D. Neither drug is appropriate for Salmonella
- E. Doxycycline is the required alternative in children

8. A patient with hepatic impairment (Child-Pugh B cirrhosis) and chronic HCV is being considered for sofosbuvir. Why does this drug require less dose caution regarding hepatic impairment compared to many other hepatically metabolized drugs?

MEDIUM

- A. Sofosbuvir is entirely renally cleared with no hepatic involvement
- B. Sofosbuvir is a prodrug triphosphorylated within hepatocytes during first-pass metabolism, activating right at its site of action in the liver — the lecture notes no dose adjustment is needed for hepatic or mild-moderate renal impairment
- C. Sofosbuvir has no described pharmacokinetic profile relevant to hepatic function
- D. Sofosbuvir is contraindicated in any degree of hepatic impairment
- E. Sofosbuvir requires a higher dose in hepatic impairment, not a lower one

9. A patient with chronic HCV is prescribed ribavirin combined with peginterferon. Pre-treatment screening reveals moderate anemia. What is the concern with proceeding?

MEDIUM

- A. Ribavirin has no hematologic effects
- B. Ribavirin causes dose-dependent hemolytic anemia and is contraindicated in existing anemia
- C. Anemia is only a concern with sofosbuvir, not ribavirin
- D. Ribavirin would correct the anemia therapeutically
- E. This concern applies only to pediatric patients

10. A patient with chronic HBV is prescribed tenofovir rather than lamivudine as initial therapy. What is the pharmacological rationale for this preference?

MEDIUM

- A. Tenofovir has a shorter half-life requiring more frequent dosing
- B. Tenofovir has low resistance and remains active against lamivudine/entecavir-resistant strains, unlike lamivudine which is associated with high rates of resistant mutations
- C. Lamivudine is more effective against HCV, making tenofovir the HBV-only choice
- D. Tenofovir is contraindicated in HBV mono-infection
- E. There is no described difference in resistance profiles between the two drugs

11. A patient with traveler's diarrhea from suspected ETEC is being treated empirically. Which antibiotic option is specifically noted as safe even in pregnancy and children under 14, unlike ciprofloxacin?

MEDIUM

- A. Ciprofloxacin
- B. Azithromycin and rifaximin
- C. Doxycycline
- D. Tetracycline
- E. None of the traveler's diarrhea antibiotics are safe in these populations

12. A patient with a complicated intra-abdominal infection (cholecystitis with risk of anaerobic involvement) is prescribed ciprofloxacin. Based on the described regimen pattern, what additional agent should typically accompany it?

MEDIUM

- A. No additional agent is needed
- B. Metronidazole, since ciprofloxacin/levofloxacin regimens for complicated intra-abdominal infections are paired with metronidazole for anaerobic coverage
- C. A second fluoroquinolone
- D. An antiviral agent
- E. Ribavirin

13. A patient starting Vosevi (sofosbuvir/velpatasvir/voxilaprevir) for HCV is also taking a hormonal contraceptive. What interaction concern applies?

MEDIUM

- A. No interaction exists between Vosevi and contraceptives
- B. Vosevi must not be combined with contraceptives per the described interaction list
- C. Contraceptives increase Vosevi's antiviral efficacy
- D. This interaction only matters in men
- E. Vosevi is specifically indicated to replace contraceptives

14. A patient with well-compensated (mild) cirrhosis and chronic HCV without cirrhosis-related complications is being considered for Vosevi. What patient population is Vosevi specifically indicated for, per the lecture?

MEDIUM

- A. Only decompensated cirrhosis patients
- B. HCV patients without cirrhosis or with mild cirrhosis specifically
- C. Only HBV patients
- D. Only patients with acute (not chronic) HCV
- E. Vosevi has no specific population restriction described

15. A patient with chronic HBV monoinfection is being started on antiviral therapy, while a separate patient has chronic HCV monoinfection. Both are being considered for interferon-alpha versus a more targeted agent. Based on the described drug-class split (interferon for both HBV/HCV; nucleos(t)ide analogues HBV-only; sofosbuvir/ribavirin/protease inhibitors HCV-specific), what is the most defensible general principle for modern first-line selection illustrated by this framework?

HARD

- A. Interferon should always be first-line for both conditions given its broader spectrum
- B. Despite interferon's broader spectrum covering both HBV and HCV, the described virus-specific agents (nucleos(t)ide analogues for HBV; sofosbuvir/ribavirin/protease inhibitors for HCV) represent more targeted mechanisms, and clinical selection typically weighs targeted, virus-specific agents against interferon's broader but less selective (and more toxic, per its described ADR profile) mechanism
- C. HBV and HCV require identical drug regimens with no meaningful distinction
- D. Nucleos(t)ide analogues are effective against both HBV and HCV despite being described as HBV-only
- E. Protease inhibitors are the first-line choice for HBV specifically

16. A researcher notes that most bacterial GI infections are "self-limited" and antibiotics are reserved for bacteremia/septicemia or debilitated patients, yet the lecture also lists specific first-line antibiotics for organisms like Salmonella, Shigella, Campylobacter, and Vibrio cholerae. How should these two pieces of information be reconciled in clinical practice?

HARD

- A. These are contradictory and cannot both be true
- B. The listed organism-specific antibiotic regimens are for cases that meet treatment criteria (e.g., severe disease, bacteremia risk, specific organisms like Shigella/cholera where treatment shortens illness/transmission) rather than implying all bacterial GI infections uniformly require antibiotics — the general "self-limited, ORT-first" principle still applies as the default for most uncomplicated bacterial gastroenteritis
- C. All bacterial GI infections must always be treated with antibiotics regardless of severity, contradicting the self-limited principle entirely
- D. Only viral GI infections are ever self-limited; all bacterial infections require antibiotics always
- E. The organism-specific regimens apply only to viral infections, not bacterial ones

17. A patient with chronic HCV is being counseled about Vosevi's drug interaction profile, which lists rifampicin, carbamazepine, phenobarbital, and phenytoin as agents that must not be combined with it because "enzyme inducers cut its efficacy." The same patient has well-controlled epilepsy on phenytoin. What is the most defensible clinical reasoning here?

HARD

- A. Phenytoin should simply be continued unchanged alongside Vosevi with no further consideration
- B. Since phenytoin is explicitly listed as an enzyme inducer that reduces Vosevi's efficacy, this combination requires either an alternative antiepileptic not on the interacting list, an alternative antiviral regimen, or close specialist coordination — simply continuing both as originally prescribed risks HCV treatment failure
- C. This interaction only matters for HBV treatment, not HCV, so no action is needed
- D. Phenytoin has no described interaction with any hepatitis antiviral
- E. Vosevi should be doubled in dose to overcome the interaction

18. A patient with cirrhosis and chronic HBV is being evaluated for antiviral therapy. The clinician is weighing entecavir against tenofovir, aware that both are nucleos(t)ide analogues with HBV-only activity. Considering the described resistance profiles, which reasoning would most appropriately guide the choice in a treatment-experienced patient with prior lamivudine exposure?

HARD

- A. Entecavir and tenofovir have identical resistance profiles, making either choice equivalent regardless of prior lamivudine exposure
- B. Since lamivudine is associated with high rates of resistant mutations, and tenofovir is specifically noted as active against lamivudine/entecavir-resistant strains, tenofovir would be the more defensible choice in a patient with prior lamivudine exposure and potential cross-resistance concern
- C. Prior lamivudine exposure has no bearing on selecting subsequent nucleos(t)ide analogues
- D. Entecavir should always be chosen over tenofovir regardless of treatment history
- E. This scenario requires switching to ribavirin, an HCV-specific drug, instead

19. A public health team is designing an oral rehydration strategy for a cholera outbreak and is deciding between the original ORS formula and the WHO low-osmolarity formula. Based on the described benefits of the low-osmolarity formula (reduced stool output, vomiting, and need for IV fluids), what is the most defensible rationale for adopting it as the outbreak-response standard?

HARD

- A. The original formula and low-osmolarity formula are functionally identical, so the choice is arbitrary
- B. The low-osmolarity WHO ORS formula has documented benefits (reduced stool output, vomiting, and IV fluid needs) over the original formula, providing an evidence-based rationale for adopting it as the standard in an outbreak setting where resource efficiency and reduced IV fluid burden matter most
- C. IV fluids should replace ORS entirely in any outbreak setting
- D. ORS composition has no bearing on clinical outcomes in cholera
- E. The original ORS formula is superior and should be used instead

20. A patient with HCV and no cirrhosis, previously treated with an interferon-based regimen that failed, is now being considered for a fixed-dose combination product blocking multiple sites on the viral RNA polymerase and protease. Which agent, and why would this multi-target approach be theoretically advantageous in a treatment-experienced, potentially resistance-harboring patient?

HARD

- A. Ribavirin alone, since it targets only the RNA-dependent RNA polymerase and has no combination benefit
- B. Vosevi (sofosbuvir + velpatasvir + voxilaprevir), which blocks 2 sites on the viral RNA polymerase (sofosbuvir, velpatasvir) plus the protease (voxilaprevir) — three proteins essential for HCV replication — theoretically raising the barrier to resistance compared to single-target agents in a treatment-experienced patient
- C. Interferon-alpha alone, since it is described as effective for both HBV and HCV
- D. Lamivudine, since it is a nucleos(t)ide analogue effective against resistant HCV strains
- E. Tenofovir, since it has the lowest resistance profile among all antivirals described

✓ Answer key

Q	Answer	Difficulty	Why
1	B	EASY	No antiviral drug exists for these; management is conservative (antipyretics, antiemetics, ORS).
2	B	EASY	Ciprofloxacin 500mg/12h for 1-3 days is the WHO first-line for Shigella.
3	B	EASY	The first step is always to stop the causative antibiotic before starting specific treatment.

4	B	EASY	Interferon-alpha is effective for both HBV and HCV, unlike the HBV-only or HCV-only specific agents.
5	C	MEDIUM	Azithromycin is noted as the appropriate alternative for pregnant women and children with cholera.
6	B	MEDIUM	The offending antibiotic is stopped first; oral metronidazole is the first-choice specific therapy, with vancomycin reserved for treatment failure.
7	B	MEDIUM	Ciprofloxacin is contraindicated in patients under 14 years; ceftriaxone is the safe alternative for Salmonella in this population.
8	B	MEDIUM	Sofosbuvir activates directly in hepatocytes during first-pass metabolism, and the lecture notes no dose adjustment is needed for hepatic or mild-moderate renal impairment.
9	B	MEDIUM	Ribavirin causes dose-dependent hemolytic anemia and is contraindicated in patients with existing anemia (also end-stage renal failure, severe heart disease, pregnancy).
10	B	MEDIUM	Lamivudine has high rates of resistant mutations, while tenofovir has low resistance and remains active against lamivudine/entecavir-resistant strains.
11	B	MEDIUM	Azithromycin and rifaximin are noted as safe in patients under 14, during breastfeeding, and in pregnancy, unlike ciprofloxacin.
12	B	MEDIUM	The regimen pattern described pairs ciprofloxacin/levofloxacin with metronidazole (each of the last three plus metronidazole) for complicated intra-abdominal infections.
13	B	MEDIUM	Vosevi's described interaction list explicitly includes contraceptives among the drugs it must not be combined with.
14	B	MEDIUM	Vosevi is described as being for HCV without cirrhosis or with mild cirrhosis.
15	B	HARD	While interferon-alpha's broader spectrum covers both HBV and HCV, its described ADR profile (flu-like symptoms, marrow suppression, depression) is more burdensome than the virus-specific targeted agents, illustrating why targeted mechanisms (nucleos(t)ide analogues for HBV; sofosbuvir-based regimens for HCV) are generally favored where applicable despite interferon's broader theoretical coverage.
16	B	HARD	The organism-specific regimens represent appropriate treatment for situations that meet clinical criteria for antibiotic therapy (specific organisms like Shigella and cholera where treatment is beneficial, or when bacteremia/severe/debilitated-patient criteria are met), while the general "self-limited/ORT-first" principle remains the overarching default for most uncomplicated bacterial GI infections — the two statements describe different clinical scenarios within the same overall framework, not a contradiction.
17	B	HARD	Since phenytoin is explicitly listed among the enzyme inducers that reduce Vosevi's efficacy, continuing both without modification risks HCV treatment failure — appropriate management requires considering an alternative antiepileptic or antiviral regimen, or specialist coordination, rather than simply continuing the original phenytoin regimen unchanged.
18	B	HARD	Since tenofovir is specifically described as active against lamivudine/entecavir-resistant strains (with low resistance overall), it is the more defensible choice in a treatment-experienced patient with prior lamivudine exposure where resistant mutations may already be present.
19	B	HARD	The lecture explicitly states the low-osmolarity WHO ORS formula reduces stool output, vomiting, and need for IV fluids compared to the original formula — directly supporting its adoption as the outbreak-response standard, particularly where resource and IV fluid capacity are constrained.
20	B	HARD	Vosevi's three-component design (blocking two sites on RNA polymerase plus the protease — three proteins essential for HCV replication) is designed to raise the barrier to resistance compared to single-target agents, which is particularly relevant in a treatment-experienced patient who may already harbor resistance-associated mutations from prior therapy.